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2026 NCE Specimen Examination

2026

MATHEMATICS

Paper 2 of 4 | Standard Form · Simultaneous Equations · Trigonometry · Probability

Time Allowed: 2 hours **Total Marks:** 100 **Candidates:** Grade 9 NCE

Index Number: _____

Formulae

Standard form: $a \times 10^n$, where $1 \leq a < 10$

Pythagoras: $a^2 + b^2 = c^2$

Area: Triangle $A = \frac{1}{2}bh$ | Circle $A = \pi r^2$

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Trigonometry: $\sin \theta = \text{opp/hyp}$ | $\cos \theta = \text{adj/hyp}$ | $\tan \theta = \text{opp/adj}$

Probability: $P(A) = n(A)/n(S)$ | $P(A \text{ or } B) = P(A) + P(B)$ for mutually exclusive events

Volume: Cuboid $V = lwh$ | Cylinder $V = \pi r^2 h$

Circumference: $C = 2\pi r$ | Sector area = $\frac{\theta}{360} \pi r^2$

Instructions to Candidates

1. Write your Index Number in the space provided above.
2. Answer ALL questions in Sections A, B and C.
3. In Section A, tick ONE box only for each question.
4. Show all working clearly in Sections B and C.
5. Marks available are shown in brackets [].
6. You may use a calculator. $\pi = 3.142$ unless otherwise stated.
7. This paper contains 28 questions totalling 100 marks.

SECTION A — Multiple Choice Questions (10 marks) [Q1–10, 1 mark each]

For each question tick ONE box only.

- 1** Which of the following is correctly written in standard form? [1]
- ☐ A) 43.7×10^{-2}
- ☐ B) 4.37×10^{-3}
- ☐ C) 0.437×10^{-4}
- ☐ D) 437×10^{-1}
- 2** The value of 6.02×10^4 as an ordinary number is: [1]
- ☐ A) 602
- ☐ B) 6020
- ☐ C) 60 200
- ☐ D) 602 000
- 3** If $x+y=10$ and $x-y=4$, then x equals: [1]
- ☐ A) 3
- ☐ B) 7
- ☐ C) 6
- ☐ D) 14
- 4** The value of $\sin 30^\circ$ is: [1]
- ☐ A) 1
- ☐ B) $\sqrt{3}/2$
- ☐ C) $1/2$
- ☐ D) $\sqrt{2}/2$
- 5** In a right-angled triangle, the hypotenuse is 10 cm and one angle is 30° . The side opposite the 30° angle is: [1]
- ☐ A) 5 cm
- ☐ B) 8.66 cm
- ☐ C) 10 cm
- ☐ D) 20 cm
- 6** The probability of getting a head when a fair coin is tossed is: [1]

- ☐ A) 0
- ☐ B) $\frac{1}{4}$
- ☐ C) $\frac{1}{2}$
- ☐ D) 1

7 Which is the smallest number?

[1]

- ☐ A) 3.2×10^{-2}
- ☐ B) 4.1×10^{-1}
- ☐ C) 8.5×10^{-1}
- ☐ D) 2.9×10^{-3}

8 Given $2x+y=7$ and $x+y=5$, the value of y is:

[1]

- ☐ A) 1
- ☐ B) 2
- ☐ C) 3
- ☐ D) 4

9 The value of $\cos 45^\circ$ is:

[1]

- ☐ A) 1
- ☐ B) $\frac{1}{2}$
- ☐ C) $\frac{\sqrt{2}}{2}$
- ☐ D) $\frac{\sqrt{3}}{2}$

10 A bag contains 3 red and 7 blue balls. The probability of drawing a red ball at random is:

[1]

- ☐ A) $\frac{3}{7}$
- ☐ B) $\frac{7}{10}$
- ☐ C) $\frac{3}{10}$
- ☐ D) $\frac{1}{3}$

SECTION B — Short Structured Questions (48 marks) [Q11–22]

Show all working. Answers without working may not receive full marks.

11 Standard Form — Conversion

[3]

- (a) Write 0.000 385 in standard form.

[1]

- (b) Write 7.4×10^5 as an ordinary number.

[1]

- (c) Write the numbers 3.1×10^4 , 2.8×10^3 and 5.6×10^4 in order from smallest to largest.

[1]

12 Standard Form — Calculations

[3]

- (a) Calculate $(3 \times 10^4) \times (2 \times 10^3)$. Give your answer in standard form.

[2]

- (b) Calculate $(8 \times 10^6) \div (4 \times 10^2)$. Give your answer in standard form.

[1]

13 Simultaneous Equations — Elimination**[4]**

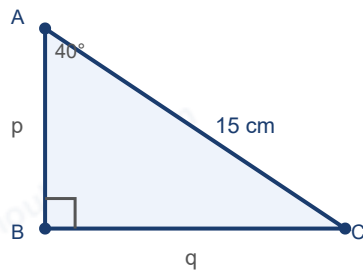
Solve the simultaneous equations:

$$3x + 2y = 16$$

$$x + 2y = 8$$

 $x =$ $y =$ **14 Trigonometry — Finding Sides****[4]**

In the right-angled triangle below, find the lengths of the unknown sides.

**(a)** Find p (the side opposite 40°).**[2]****(b)** Find q (the side adjacent to 40°).**[2]**



15 Probability — Simple Events

[4]

A spinner is divided into 8 equal sectors numbered 1 to 8.

(a) Find $P(\text{even number})$.

[1]

(b) Find $P(\text{prime number})$. [2, 3, 5, 7 are prime.]

[1]

(c) Find $P(\text{number greater than 5})$.

[1]

(d) Find $P(9)$.

[1]

16 Standard Form — Real-World Context

[4]

(a) The distance from Earth to the Sun is approximately 150 000 000 km. Write this in standard form.

[1]

(b) A red blood cell has diameter 0.000 007 m. Write this in standard form.

[1]

- (c) The speed of light is 3×10^8 m/s. How far does light travel in 500 seconds? Give your answer in standard form. [2]

17 Simultaneous Equations — Word Problem

[4]

Two numbers x and y satisfy: their sum is 20 and their difference is 6.

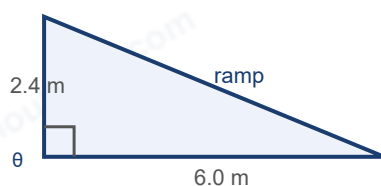
- (a) Write two equations in x and y . [2]

- (b) Solve to find x and y . [2]

18 Trigonometry — Finding an Angle

[5]

A ramp rises 2.4 m vertically over a horizontal distance of 6.0 m.



- (a) Write down which trigonometric ratio connects θ , the rise and the horizontal distance. [1]

- (b) Calculate the angle θ that the ramp makes with the horizontal. [2]

- (c) Calculate the length of the ramp. [2]

19 Probability — Complementary and Combined Events [4]

- (a) The probability that it rains on any day in June is 0.35. Find the probability that it does NOT rain on a randomly chosen day. [1]

- (b) A fair die is rolled once. Find $P(\text{score is 2 or 5})$. [2]

- (c) Events A and B are mutually exclusive. $P(A)=0.3$ and $P(B)=0.45$. Find $P(A \text{ or } B)$. [1]

20 Standard Form — Mixed Calculations

[4]

- (a) Calculate $(5.4 \times 10^6) + (2.6 \times 10^6)$. Give your answer in standard form. [2]

- (b) Calculate $(6 \times 10^3)^2$. Give your answer in standard form. [2]

21 Simultaneous Equations — Substitution Method

[5]

Solve using substitution:

$$\begin{aligned}y &= 2x - 1 \\ 3x + 2y &= 20\end{aligned}$$

x =

y =

22 Probability — Two Events

[4]

A bag contains 5 red, 3 green and 2 yellow balls. A ball is drawn at random.

(a) Find $P(\text{red})$.

[1]

(b) Find $P(\text{not green})$.

[1]

(c) Two balls are drawn without replacement. Find $P(\text{both red})$.

[2]

SECTION C — Long Structured Questions (42 marks) [Q23–28, 7 marks each]

Show ALL working. Marks may be awarded for correct method even if the final answer is wrong.

23 Standard Form — Comprehensive

[7]

- (a) The mass of a proton is 1.67×10^{-27} kg and the mass of an electron is 9.11×10^{-31} kg. How many times heavier is a proton than an electron? Give your answer in standard form correct to 3 significant figures. [3]

- (b) Calculate $(4.8 \times 10^5) \div (1.6 \times 10^{-3})$. Give your answer in standard form. [2]

- (c) The population of a city is 2.4×10^6 . Each person uses an average of 150 litres of water per day. Find the total daily water usage in standard form. [2]

24 Simultaneous Equations — Extended

[7]

(a) Solve simultaneously: $5x+3y=29$ and $2x+3y=17$.

[4]

x =

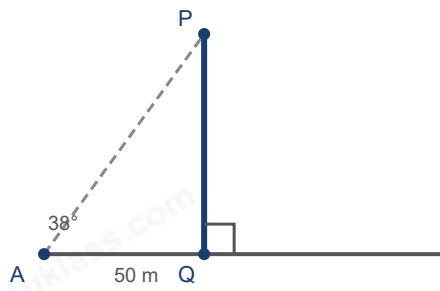
y =

(b) A school sells adult tickets at Rs x each and child tickets at Rs y each. Three adult and five child tickets cost Rs 430. Two adult and two child tickets cost Rs 240. Form two equations and solve to find x and y . [3]

25 Trigonometry — Practical Application

[7]

A vertical tower PQ stands on horizontal ground. From a point A on the ground, 50 m from the base Q of the tower, the angle of elevation of the top P is 38° .



- (a) Calculate the height PQ of the tower. [3]

- (b) Calculate the distance AP (the line of sight). [2]

- (c) From a point B, also on the ground, the angle of elevation of P is 55° . Calculate the distance BQ. [2]

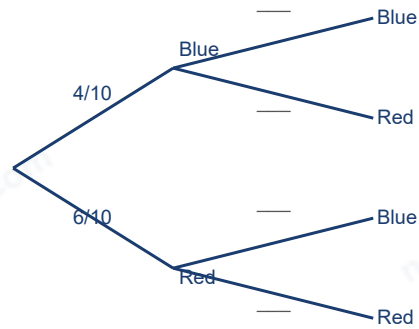
26 Probability — Tree Diagram

A box contains 4 blue and 6 red pens. A pen is drawn at random, its

[7]

colour recorded, and it is then replaced. A second pen is then drawn.

- (a) Complete the tree diagram below. [3]



- (b) Find $P(\text{both pens are the same colour})$. [2]

- (c) Find $P(\text{at least one red pen})$. [2]

27 Simultaneous Equations — Applications

[7]

- (a) Solve simultaneously: $4x - 3y = 1$ and $2x + y = 7$. [4]

x =

y =

- (b) The perimeter of a rectangle is 34 cm. The length is 3 cm more than twice the width. Find the dimensions of the rectangle. Show all working. [3]

28 Trigonometry — Extended Problem

[7]

In triangle ABC, angle B = 90° . AB = 8 cm and BC = 15 cm.

- (a) Calculate AC. [2]

- (b) Calculate angle BAC (give your answer to 1 decimal place). [2]

- (c) D is the midpoint of AC. Calculate the length BD using Pythagoras' theorem (hint: find AD first). [3]



Mark Scheme — Paper 2 (Standard Form · Simultaneous Equations · Trigonometry · Probability) 2026

Equivalent correct methods receive full credit. M=method mark; A=accuracy mark. Allow ± 0.1 on trig answers unless specified. Use calculator values throughout.

Q	Answer / Working	Marks	Notes
SECTION A			
1	B — 4.37×10^3 (coefficient between 1 and 10)	1	
2	C — 60 200	1	
3	B — $x=7$ (adding: $2x=14$, $x=7$)	1	
4	C — $1/2$	1	
5	A — 5 cm ($\sin 30^\circ \times 10 = 0.5 \times 10 = 5$)	1	
6	C — $1/2$	1	
7	C — $8.5 \times 10^{-1} = 0.85$ (smallest)	1	
8	C — $y=3$ (subtract: $x=2$; $y=5-2=3$)	1	
9	C — $\sqrt{2}/2$ (≈ 0.707)	1	
10	C — $3/10$	1	
SECTION B			
11(a)	3.85×10^{-4}	1	
11(b)	740 000	1	
11(c)	$2.8 \times 10^3 < 3.1 \times 10^4 < 5.6 \times 10^4$	1	
12(a)	$3 \times 2 = 6$; $10^4 \times 10^3 = 10^7$; 6×10^7	2	M1; A1
12(b)	$8/4 = 2$; $10^{6-2} = 10^4$; 2×10^4	1	
13	Subtract: $2x=8$, $x=4$; substituting: $4+2y=8$, $y=2$	4	M1 eliminating; M1 value of x; M1 substituting; A1 y
14(a)	$p = 15 \times \sin 40^\circ = 15 \times 0.6428 = \mathbf{9.64 \text{ cm}}$	2	M1 correct trig ratio; A1
14(b)	$q = 15 \times \cos 40^\circ = 15 \times 0.7660 = \mathbf{11.49 \text{ cm}}$	2	M1; A1
15(a)	$4/8 = \mathbf{1/2}$	1	
15(b)	$4/8 = \mathbf{1/2}$	1	
15(c)	$3/8$ (6,7,8 $\rightarrow$ 3 numbers)	1	

15(d)	0 (9 is not on the spinner)	1	
16(a)	1.5×10^8 km	1	
16(b)	7×10^{-6} m	1	
16(c)	$3 \times 10^8 \times 500 = 3 \times 10^8 \times 5 \times 10^2 = \mathbf{1.5 \times 10^{11} \text{ m}}$	2	M1; A1
17(a)	$x+y=20$ and $x-y=6$ (or $y-x=6$)	2	M1 each equation
17(b)	Add: $2x=26$, $x=13$; $y=20-13=7$	2	M1; A1
18(a)	$\tan\theta = \text{opp/adj}$ (or $\tan\theta = 2.4/6.0$)	1	
18(b)	$\tan\theta = 2.4/6.0 = 0.4$; $\theta = \tan^{-1}(0.4) = \mathbf{21.8^\circ}$	2	M1; A1
18(c)	$\text{Ramp}^2 = 2.4^2 + 6.0^2 = 5.76 + 36 = 41.76$; ramp = 6.46 m (or: $2.4/\sin 21.8^\circ$)	2	M1; A1
19(a)	$1 - 0.35 = \mathbf{0.65}$	1	
19(b)	$P(2) + P(5) = 1/6 + 1/6 = \mathbf{2/6 = 1/3}$	2	M1; A1
19(c)	$0.3 + 0.45 = \mathbf{0.75}$	1	
20(a)	$5.4 + 2.6 = 8.0$; 8.0×10^6	2	M1; A1
20(b)	$36 \times 10^6 = \mathbf{3.6 \times 10^7}$	2	M1; A1
21	Sub $y=2x-1$: $3x+2(2x-1)=20$; $3x+4x-2=20$; $7x=22$; $x=22/7 \approx 3.14$; $y=2(22/7)-1=44/7-7/7=\mathbf{37/7 \approx 5.29}$. Accept fractions or decimals.	5	M1 substituting; M1 expanding; M1 solving x; A1 x; A1 y
22(a)	$5/10 = \mathbf{1/2}$	1	
22(b)	$P(\text{not green}) = 1 - 3/10 = \mathbf{7/10}$	1	
22(c)	$P(1\text{st red})=5/10$; $P(2\text{nd red} 1\text{st red})=4/9$; $P(\text{both red})=5/10 \times 4/9 = 20/90 = \mathbf{2/9}$	2	M1 product; A1
SECTION C			
23(a)	$1.67 \times 10^{-27} \div 9.11 \times 10^{-31} =$ $(1.67/9.11) \times 10^{-27-(-31)} = 0.1833 \times 10^4 = \mathbf{1.83 \times 10^3}$ (to 3 s.f.)	3	M1 dividing coefficients; M1 index; A1
23(b)	$(4.8/1.6) \times 10^{5-(-3)} = 3 \times 10^8 = \mathbf{3.0 \times 10^8}$	2	M1; A1
23(c)	$2.4 \times 10^6 \times 150 = 2.4 \times 10^6 \times 1.5 \times 10^2 = 3.6 \times 10^8$ litres	2	M1; A1
24(a)	Subtract: $3x=12$, $x=4$; $3(4)+2y=29$, $2y=17$, $y=8.5$	4	M1 eliminating; M1 x; M1 sub; A1 y
24(b)	$3x+5y=430$ and $2x+2y=240 \rightarrow x+y=120$; from sub: $x=120-y$; $3(120-y)+5y=430$; $360+2y=430$; $y=35$; $x=85$. Adult=Rs 85, Child=Rs 35.	3	M1 equations; M1 solving; A1
25(a)	$PQ = 50 \times \tan 38^\circ = 50 \times 0.7813 = \mathbf{39.06 \text{ m}}$	3	M1 tan ratio; M1 substituting; A1

25(b)	$AP = 50 / \cos 38^\circ = 50 / 0.7880 = \mathbf{63.46 \text{ m}}$ (or $\sqrt{50^2 + 39.06^2}$)	2	M1; A1
25(c)	$BQ = PQ / \tan 55^\circ = 39.06 / 1.4281 = \mathbf{27.35 \text{ m}}$	2	M1; A1
26(a)	Second draw branches: Blue B: 4/10, Red B: 6/10; Blue R: 4/10, Red R: 6/10	3	1 mark per correct pair of branches
26(b)	$P(BB) = (4/10)(4/10) = 16/100$; $P(RR) = (6/10)(6/10) = 36/100$; Total = $\mathbf{52/100 = 13/25}$	2	M1; A1
26(c)	$P(\text{at least 1 red}) = 1 - P(\text{no red}) = 1 - P(BB) = 1 - 16/100 = \mathbf{84/100 = 21/25}$	2	M1 complement; A1
27(a)	From $2x + y = 7$: $y = 7 - 2x$; sub: $4x - 3(7 - 2x) = 1$; $10x = 22$; $\mathbf{x = 2.2}$; $y = 7 - 4.4 = \mathbf{2.6}$	4	M1 substituting; M1 expanding; M1 x; A1 y
27(b)	Let width = w ; length = $2w + 3$; $2(w + 2w + 3) = 34$; $2(3w + 3) = 34$; $6w + 6 = 34$; $6w = 28$; $\mathbf{w = 14/3 \approx 4.67 \text{ cm}}$; length = $2(14/3) + 3 = 37/3 \approx \mathbf{12.33 \text{ cm}}$. Check: $2(14/3 + 37/3) = 2(51/3) = 34 \checkmark$	3	M1 equations; M1 solving; A1 both dimensions
28(a)	$AC^2 = 8^2 + 15^2 = 64 + 225 = 289$; $\mathbf{AC = 17 \text{ cm}}$	2	M1; A1
28(b)	$\tan(BAC) = BC/AB = 15/8 = 1.875$; $BAC = \tan^{-1}(1.875) = \mathbf{61.9^\circ}$	2	M1; A1
28(c)	$AD = AC/2 = 17/2 = 8.5 \text{ cm}$ (D is midpoint); in a right triangle, the median to the hypotenuse equals half the hypotenuse: $BD = AC/2 = \mathbf{8.5 \text{ cm}}$.	3	M1 identifying median property or using coordinates; A1; A1
TOTAL		100	